2020A F=ma Exam: Problem 6

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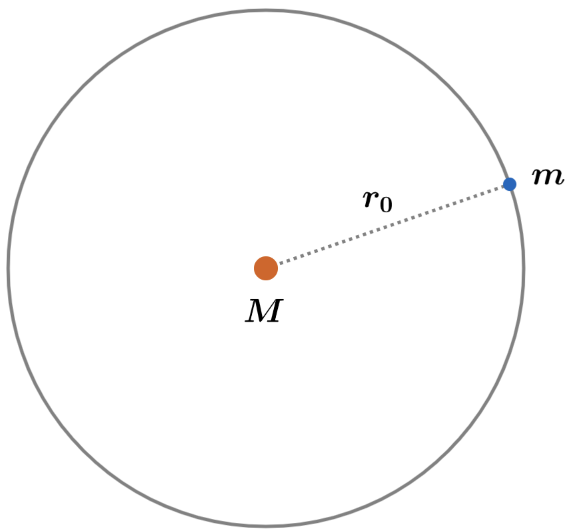

Since the star is losing its mass, the energy of the system isn't conserved. Instead, the angular momentum is conserved since there is no external torque on the system. Choosing the axis to be at the star,

$$L = mvr$$

For circular orbits, we have

$$\frac{GMm}{r^2} = \frac{mv^2}{r}$$
$$v = \sqrt{\frac{GM}{r}}$$

so

$$L \propto \sqrt{Mr}$$
$$M \propto \frac{1}{r}$$

We have

$$\frac{r_0}{r_f} = \frac{M_f}{M_0} = 0.99$$
$$r_f = 1.01 r_0$$

so the answer is $\boxed{\text{B}}$.